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1 =====Q1=====
2 #include <stdio.h>
3 #include <unistd.h>
4
5 int main(){
6     //factorial-----
7     int child1, child2;
8     int n,i,j,k;
9     printf("enter a num:");
10    scanf("%d",&n );
11    child1=fork();
12    if(child1==0){
13        int fact=1;
14        for(i=1;i<=n;i++){
15            fact=fact*i;
16        }
17
18        printf("the factorial of %d is %d\n",n,fact);
19
20    //fibonacci-----
21    int child5=fork();
22    if(child5==0){
23        int a=0, b=1, c=0;
24        for(j=1;j<=n;j++){
25            a=b+c;
26            b=c;
27            c=a;
28        }
29    }
30    printf("the fibonacci is %d\n",a);
31
32    }
33
34    // 2 to the power-----
35    else{
36        child2=fork();
37        if(child2==0){
38            int s=1;
39            if(n==0){
40                s=1;
41            }
42            else{
43                for(k=0; k<n; k++){
44                    s*=2;
45                }
46            }
47            printf("2 to the power is %d\n",s);
48        }
49    }
50
51    //square-----
52    int child3=fork();
53    if(child3==0){
54        int r= n*n;
55        printf("square of %d is %d\n",n,r);
56    }
57
58 }
59
60 =====Q2=====
61
62 #include <stdio.h>
63 #include <unistd.h>
64
65 int main(){
66     int n;
67     printf("Enter a number:");
68     scanf("%d",&n);
69     int c=fork();

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70     if(c==0){
71         printf("%d * 2 = %d\n",n,n*2);
72     }
73     else{
74         printf("%d / 2 = %d\n",n,n/2);
75     }
76     return 0;
77 }
78
79

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80     Enter a number:22
81     22 / 2 = 11
82     22 * 2 = 44
83

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84     =====Q3=====
85

```

```

86     #include <stdio.h>
87     #include <unistd.h>
88
89     int main(){
90         int i=0;
91         pid_t childpid=fork();
92         if(childpid==-1){
93             printf("faield to fork");
94             return -1;
95         }
96         else if(childpid==0){
97             for(i=0;i<5;i++){
98                 printf("child\n");
99             }
100         }
101         else{
102             for(i=0;i<5;i++){
103                 printf("parent\n");
104             }
105         }
106         return 0;
107     }
108
109

```

```

110
111     parent
112     parent
113     parent
114     parent
115     parent
116     child
117     child
118     child
119     child
120     child
121

```

```

122     =====Q4=====
123

```

```

124     #include <stdio.h>
125     #include <unistd.h>
126     int main(){
127         int a,b;
128         printf("Enter  number1:");
129         scanf("%d",&a);
130         printf("Enter  number2:");
131         scanf("%d",&b);
132
133         int c=fork();
134         if(c==0){
135             printf("%d + %d = %d\n",a,b,a+b);
136         }
137         else{
138             int d=fork();

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139         if(d==0){
140             printf("%d - %d = %d\n",a,b,a-b);
141         }
142         else{
143             int e=fork();
144             if(e==0){
145                 printf("%d * %d = %d\n",a,b,a*b);
146             }
147             else{
148                 int f=fork();
149                 if(f==0){
150                     printf("%d / %d = %d\n",a,b,a/b);
151                 }
152             }
153         }
154     }
155     return 0;
156 }

```

=====Q5=====

```

160 #include <stdio.h>
161 #include <unistd.h>
162 #include <sys/msg.h>
163 #include <string.h>
164 #include <stdlib.h>
165 #define MAXSIZE 128
166 #define MSGKEY 1111
167 int main(){
168     int msgpid;
169
170     struct person{ //create structure
171         int age;
172         char name[100]; //declare as array in char data type
173         char regno[100]; // same as above
174     };
175
176     struct person p1,p2; // p1 for centre, p2 for reciever
177     int f;
178     f=fork();
179     if(f==0){
180         int msgpid;
181         msgpid = msgget (MSGKEY, IPC_CREAT|0666);
182         msgrcv(msgpid,&p2,1024,0,0);
183         printf("Recieved name:%s\n",p2.name);
184         printf("Recieved regno:%s\n",p2.regno);
185         printf("Recieved age:%d\n",p2.age);
186         msgctl(msgpid,IPC_RMID,0);
187     }
188     else{
189         msgpid = msgget (MSGKEY, IPC_CREAT|0666);
190         printf("Enter the name: ");
191         scanf("%s",&p1.name);
192         printf("Enter the regno: ");
193         scanf("%s",&p1.regno);
194         printf("Enter the age: ");
195         scanf("%d",&p1.age);
196
197         int buflen=sizeof(p1)+1;
198
199         if(msgsnd(msgpid,&p1,buflen,IPC_NOWAIT)<0){
200             printf("Message not send\n");
201         }
202         else{
203             printf("Message send\n");
204             exit(0);
205         }
206     }
207 }

```